

Automated Integration and Intelligent Harmonization of Multi-Source Geospatial Data for Urban Land Record Management

KSHETRA — research summary and evidence base

Team NovaX · Smart India Hackathon 2026 · Problem Statement 26013

This document states what the published literature establishes, what it leaves open, and which of those open problems our prototype addresses. Every performance figure quoted for KSHETRA was measured by a harness in the repository, not estimated.

1. The problem, quantified

Urban land administration draws on drone orthoimagery, DSM/DTM surfaces, scanned cadastral sheets, revenue registers, municipal GIS, utility networks and GNSS control. These disagree in coordinate system, position, geometry, attribute schema, vintage and completeness. Reconciling them is presently manual GIS work.

The scale of the mismatch is documented. Sengupta, Bhattacharya and Mukherjee measured registration error across 310 cadastral sheets in West Bengal and found a modal RMSE of **3–4 m** [5]. The Survey of India specification for the NAKSHA programme requires orthoimagery accurate to **$RMSE(x,y) \leq 10\text{ cm}$** and DSM/DTM to **$RMSE(z) \leq 15\text{ cm}$** [6]. Legacy geometry is therefore wrong by roughly **thirty to forty times** the tolerance of the imagery it must be reconciled against, which is why naive overlay produces nothing usable.

The administrative position is consistent with this. Under DILRMP, only **49.10%** of villages hold geo-referenced cadastral maps [7].

2. What the field says remains unsolved

Chen, Nazeer, Lee and Wong published a systematic review of artificial intelligence in cadastre in March 2026 [1]. It concludes that deep learning has largely solved automated boundary extraction, that natural-language methods now handle non-spatial records, and that change detection is established. It then names what has not been solved:

“...challenges remain, including differences in multi-temporal data processing, spatial semantic ambiguity, and the lack of large-scale, high-quality annotated data. Future research can focus on improving model generalization, advancing cross-modal data fusion...” [1]

Three of those four are directly relevant here, and they are the axes on which this project is built: **the absence of annotated ground truth**, **cross-modal fusion**, and **generalisation beyond the area a method was tuned on**.

3. Prior art, and what each leaves open

3.1 Alignment under noisy supervision — Girard, Charpiat & Tarabalka [2]

The closest published statement of our core problem. The authors align misregistered cadastral polygons to imagery when no correct annotation exists, using a multi-resolution U-Net that predicts a displacement field, trained in repeated rounds that re-correct the annotations between rounds. Alignment error fell by more than a factor of three by the second round, with the third adding nothing [2].

Two findings matter for us. First, their training data is built by applying **known random deformations** to annotations and learning to invert them — the same principle our damage harness uses, and the reason we

consider that approach established rather than improvised. Second, they report that a perfect score is unattainable because the reference annotations are themselves ambiguous, and that their model learns only **smooth** displacement fields, so adjoining buildings needing discontinuous correction fail [2].

Our Gaussian-process error field shares that smoothness assumption. We treat it as a stated limitation rather than a solved problem, and it is why our uncertainty model reports an irreducible floor (0.75 m here) that no amount of survey control removes.

3.2 Cadastral parcel adjustment with SAM and ICP — Suwardhi et al. [3]

The closest operational system. UAV orthophotos at 5 cm GSD are segmented by the Segment Anything Model without manual prompting; displacement vectors come from ICP; adjustment proceeds hierarchically from rigid block (LS1) to scaled block (LS2) to individual parcel (LS3). Evaluated over two urban villages in Cimahi, Indonesia — Karangmekar, 81 blocks and 3,227 parcels, and Baros, 96 blocks and 2,971 parcels [3].

What it leaves open is instructive. Because no correct geometry exists for those areas either, the authors evaluate **by proxy**: they count how many times cadastral polygons split the SAM-derived boundaries, and treat fewer splits as better alignment [3]. That measures internal consistency, not accuracy. The work also handles a single source, does not infer attribute schema, carries no calibrated confidence, and does not address where to survey next.

3.3 Boundary revision by deep learning — Fetai, Grigillo & Lisec [4]

A modified CNN detects visible land boundaries from image-based mapping and uses them to revise existing cadastral records [4]. This is the upstream of our pipeline. We deliberately do not rebuild it: feature extraction is a solved and well-served problem, and the problem statement asks for integration of extracted features, not for another extractor.

3.4 Method the field imports from elsewhere

Our survey-planning component rests on sensor placement in Gaussian processes, where the objective is monotone submodular and greedy selection is therefore within a factor $(1 - 1/e)$ of optimal [8]. That result is standard in machine learning and, as far as we found, has not been applied to cadastral ground-control planning.

4. Position

The components are not novel individually, and we do not claim them. Segmentation, coordinate transformation, topology repair, change detection and human review all exist. The contribution is in three places, each answering one of the gaps the review names.

Manufactured ground truth, so accuracy is measured rather than asserted

Both [2] and [3] are limited by the same thing: no correct geometry exists to score against, so one reports relative improvement and the other a proxy. We generate a clean synthetic cadastre and damage a copy of it through eleven stages whose parameters are known — misregistration, smooth warp, vertex jitter, sliver and overlap injection, self-intersection, schema drift, transliteration corruption, unit drift, block renumbering, and split/merge/deletion. Recovery is then scored against the key. This addresses the review's "lack of large-scale, high-quality annotated data" [1] by constructing it, and it follows the deformation-injection principle of [2].

Calibrated confidence rather than a weighted score

A weighted sum of similarity terms is not a probability and cannot be checked. We score candidate pairs with gradient-boosted trees over 29 features, then map the output through isotonic regression fitted on held-out data, so a stated probability corresponds to an observed frequency. Expected calibration error is **0.00009**. This is what allows an explicit triage threshold instead of an arbitrary cut-off.

Closing the loop to the field

The NAKSHA programme's own 2026 progress review identifies ground truthing as its most delayed component [9]. Because every parcel carries a calibrated uncertainty, the system computes which locations to survey next by greedy submodular maximisation over the posterior variance of a fitted GP error field [8], and reports the expected reduction. The recommendation is then validated: the selected control is actually applied, georeferencing re-run, and the achieved error compared against both the prediction and against random placement.

5. Measured results

Trained on one synthetic city, evaluated on a second with a different layout and a different damage profile. Full pipeline: 45.6 s on one core.

Stage	Measure	Result
Georeferencing	positional RMSE	8.4529 m → 3.127 m
Schema inference	columns mapped; area unit	7/7; bigha at 0.9672
Blocking	true pairs retained	1.0
Matching	F1 / ROC AUC	0.9993 / 1.0
Matching	share of decision from geometry	95.6%
Confidence	expected calibration error	0.00009
Topology	total errors	6168 → 215
Topology	overlapping parcel pairs	5038 → 0
Topology	doubly-claimed land	18038.6 → 0.0 m ²
Restraint	parcel-sized gaps refused	69 of 73 deleted
Change	new / demolished / extended / heightened	31 / 28 / 70 / 59
Encroachment	sites on public land	9, 251.8 m ²
Targeting	predicted / achieved / random	1.4479 / 1.3177 / 2.0567 m

The geometric share is reported because it is a falsifiable claim about mechanism. An early version of this model scored F1 0.9955 by learning to join on khasra number and owner name, with IoU contributing 0.24% of the decision; it would have been useless in a jurisdiction that rennumbers on resurvey. The harness now rennumbers 45% of blocks and changes 12% of owners, and the test asserts a geometric share above 70%.

6. What we do not claim

- The cadastral parcel layer is **synthetic**, derived from OpenStreetMap block structure, and labelled as such throughout the product and its exports. No bulk Indian urban cadastre is publicly available. Building footprints, the road network and the terrain are real.
- Encroachment cases are **deliberately injected**, because the generator never places structures on public land; without injection there would be nothing to detect and the claim would be untested.
- The survey-plan prediction is **calibrated, not exact**. A scale factor measured on one city transfers to a held-out city within roughly 12% and errs conservative.
- Per-parcel digitising noise is **irreducible** by survey control; the system reports that floor instead of promising accuracy below it.
- The system detects and evidences change. It does **not** produce a mutation record, which is a statutory act requiring notice and hearing and whose form varies by state.

References

[1] J. Chen, M. Nazeer, B. S. Lee and M. S. Wong. *Artificial Intelligence in Cadastre: A Systematic Review of Methods*,

Applications, and Trends. Land, 15(3):411, 2026. doi:10.3390/land15030411.

[2] N. Girard, G. Charpiat and Y. Tarabalka. *Noisy Supervision for Correcting Misaligned Cadaster Maps Without Perfect Ground Truth Data*. IGARSS 2019, IEEE International Geoscience and Remote Sensing Symposium. Inria / LuxCarta.

[3] D. Suwardhi, M. Ihsan, R. Widyastuti, A. H. U. Mukminin, B. Akbar, S. K. Pasaribu, I. P. Satwika, S. L. Nurmaulia and A. Hernandi. *An Automated Framework for Cadastral Parcel Adjustment Using UAV Orthophotos, SAM, and ICP*. ISPRS Archives, XLVIII-2/W11-2025:277–284, 2025. Institut Teknologi Bandung.

[4] B. Fetai, D. Grigillo and A. Lisec. *Revising Cadastral Data on Land Boundaries Using Deep Learning in Image-Based Mapping*. ISPRS Int. J. Geo-Inf., 11(5):298, 2022. doi:10.3390/ijgi11050298.

[5] A. Sengupta et al. Measured registration error across 310 cadastral sheets, West Bengal. Survey Review, 2016. Modal RMSE 3–4 m.

[6] Survey of India. Technical circular T-260/1147-Project (NAKSHA), 10 February 2025. ORI at 5 cm GSD, $RMSE(x,y) \leq 10$ cm; DSM/DTM $RMSE(z) \leq 15$ cm; CORS control better than 5 cm; UTM projection on WGS 84.

[7] Department of Land Resources, Ministry of Rural Development. Digital India Land Records Modernisation Programme — geo-referenced cadastral map coverage, 49.10% of villages.

[8] A. Krause, A. Singh and C. Guestrin. *Near-Optimal Sensor Placements in Gaussian Processes: Theory, Efficient Algorithms and Empirical Studies*. Journal of Machine Learning Research, 9:235–284, 2008.

[9] K. Satyarthi et al. Progress review of the NAKSHA programme. Frontiers in Sustainable Cities, 2026. Identifies ground truthing as the programme's most delayed component and recommends automated GeoAI-driven updating mechanisms.

Sources [1]–[4] and [8] were read in full or in the relevant part. [5], [6], [7] and [9] are quoted from a sourced research dossier compiled for this project; figures attributed to them should be re-checked against the primary document before publication.